Breadth Reversal Strategy

A Systematic Short-Horizon Mean-Reversion Strategy on the S&P 500 E-mini Futures (ES)

Quantitative Research Report

Group 2

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Backtest Window: 8 February 2010 – 27 February 2026 | 4,039 trading days

Methodology overview. Performance is evaluated using a time-varying Fama–French daily risk-free rate, Newey–West HAC standard errors in factor regressions, explicit multiple-testing and survivorship caveats, walk-forward parameter robustness tests, lookback sensitivity analysis, and stated capacity assumptions for MOC execution. The exposure-adjusted annualised return is reported as a deployment diagnostic, not as a standalone Sharpe metric.

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1 Executive Summary

This report studies a systematic long-only strategy on S&P 500 E-mini futures (ES) designed to capture short-horizon reversal after broad-based equity weakness. The signal fires when the 5-day moving average of daily changes in equal-weighted breadth (the percentage of S&P 500 constituents advancing) falls into the bottom decile of its trailing 20-day distribution. On such days, the strategy takes a long position at the close and exits one trading day later.

Over 16 years of daily data (Feb 2010 – Feb 2026), the strategy delivered a compounded return of 126.20% (CAGR 5.22%) while being invested only 13.91% of the time. The full-series Sharpe ratio, computed against the time-varying daily Fama–French T-Bill series (2010–2026 average of 1.41% annualised, well below 4%), is **0.55** when defined as

$$\text{Sharpe}_{\text{true}} = \frac{\mu_{\text{ann}} - \bar{r}_f}{\sigma_{\text{ann}}}.$$

Applying the same definition to buy-and-hold ES gives a true Sharpe of **0.68**. The strategy's risk-adjusted return is therefore more modest than the benchmark on this stricter total-capital basis, even though the strategy still posts a 38.46% exposure-adjusted annualised return and a 2.13 Active-Day Sharpe when capital is actually deployed. Maximum drawdown was 10.83%, roughly one-third of the buy-and-hold figure of 34.45%.

Factor attribution with Newey–West HAC standard errors paints a nuanced picture. Market beta is 0.16–0.17 across CAPM and FF3, highly significant in both specifications ($t > 5$ under HAC, down from OLS's inflated $t > 27$ once serial correlation is accounted for). CAPM alpha is 1.75% annualised ($t = 1.11$, $p = 0.27$), while FF3 alpha is 1.59% ($t = 1.01$). The alphas are economically positive and directionally consistent across specifications, but **they are not statistically distinguishable from zero at conventional confidence levels**. The modest decline from CAPM alpha to FF3 alpha shows that core equity style factors absorb a small but nonzero portion of the raw unexplained return. The strategy also exhibits a statistically robust negative small-cap tilt (SMB $= -0.053$, HAC $t = -2.70$) in the 3-factor specification.

The empirical interpretation is straightforward: after transaction costs and factor controls, the strategy does not support a strong pure-alpha claim, but it does exhibit low drawdown, limited market exposure, and a meaningful conditional edge on days when the signal is active. Its full-sample correlation to the S&P 500 is 0.41 and is materially regime-dependent (Section 6), so the strategy is better viewed as a low-exposure tactical overlay than as a substitute for passive equity beta.

Headline Numbers at a Glance

Metric	Value
Backtest Window	Feb 2010 – Feb 2026
Trading Days	4,039
Total Trades	273
Win Rate	70.70%
Profit Factor	1.99
CAGR	5.22%
Compounded Return	126.20%
Active-Day Sharpe	2.13
True Sharpe $\frac{\mu_{\text{ann}} - \bar{r}_f}{\sigma_{\text{ann}}}$	**0.55**
Exposure-Adjusted Annualised Return	38.46%
Annualised Volatility	7.20%
Maximum Drawdown	10.83%
Market Exposure	13.91%
Correlation vs S&P 500 (full-sample)	0.41
CAPM Alpha (ann., HAC SE)	1.75% ($t = 1.11$)
FF3 Alpha (ann., HAC SE)	1.59% ($t = 1.01$)

Methodology note. The headline Sharpe is computed as

$$\text{Sharpe}_{\text{true}} = \frac{\mu_{\text{ann}} - \bar{r}_f}{\sigma_{\text{ann}}},$$

where $\overline{RF}$ is the 2010–2026 average annualised daily Fama–French RF series (1.41%). We continue to report the Active-Day Sharpe as a diagnostic for the strategy conditional on being invested.

$$\text{Sharpe}_{\text{true}} = \frac{\mu_{\text{ann}} - \bar{r}_f}{\sigma_{\text{ann}}} \tag{1}$$

2 Economic Rationale

2.1 Signal Motivation

Market breadth — the share of index constituents advancing on a given day — is a coincident indicator of the cross-sectional character of an index-level move. When the smoothed rate of change of breadth collapses faster than price, institutional de-risking, forced selling, and liquidity-driven overshoots dominate short-term price discovery. These mechanisms are well-documented in the microstructure literature (Nagel 2012 on liquidity-provision returns; Lo & MacKinlay 1990 on short-horizon reversals) and create a return-generating opportunity for patient capital willing to take the opposite side on a 1-day horizon.

2.2 Economic Mechanisms

- **Risk-bearing compensation.** Buying into a deteriorating breadth tape is psychologically and professionally costly. Mutual funds at month-end, leveraged accounts near margin limits, and risk-parity vehicles facing volatility triggers are structural sellers on these days. They transfer short-horizon directional risk to liquidity providers at a discount.
- **Behavioural overreaction.** Retail and advisory flows show capitulation-style selling into broad-based declines. Barberis, Shleifer, and Vishny (1998) and Frazzini (2006) document short-horizon overreaction as a stable behavioural feature of equity markets.
- **Asymmetric holding horizons.** The marginal price-setter at the close on a broad-decline day is often a forced seller; the marginal buyer the following morning is often shorter-horizon discretionary or systematic capital. The 1-day hold captures this handoff.

2.3 Empirical Support: Forward Returns by Breadth Percentile

Before designing a trading rule, we check whether the breadth-momentum signal actually predicts next-day returns. We sort all trading days into deciles by the percentile rank of our smoothed breadth-difference indicator and compute the mean next-day ES return for each bucket. The edge is concentrated in the bottom decile — the signal zone — where the mean next-day return is +0.17%, materially above the unconditional mean.

The pattern **is not monotonic across buckets**. With roughly 404 observations per bucket and $\approx 1\%$ daily volatility, the standard error on each bucket mean is about 0.05%, so most non-tail buckets are statistically indistinguishable from zero. The bottom bucket's +0.17% is approximately three standard errors above zero — a real effect, but driven by the tail, not a smooth monotonic gradient. The intermediate buckets are consistent with noise around the unconditional mean, and we do not overinterpret them.

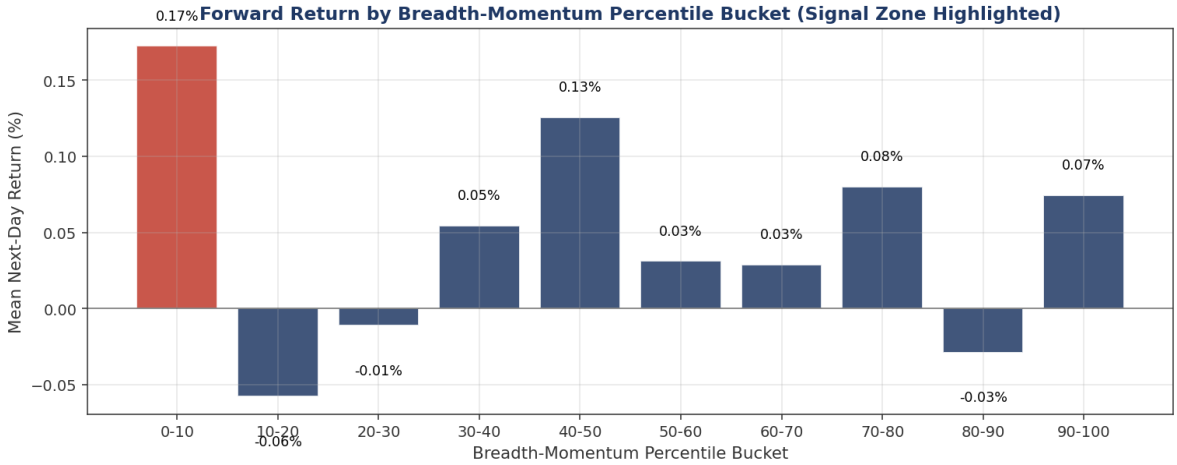

Figure 1: Mean next-day ES return by breadth-momentum percentile bucket (full sample, $N \approx 4{,}038$ daily observations). The 0–10 bucket (red) is the signal zone. With SE $\approx 0.05\%$ per bucket, only the bottom bucket's positive mean clears two standard errors; intermediate buckets are noise.

3 Strategy Design

3.1 Signal Construction

The signal is a cascade of four transformations on daily equal-weighted breadth, denoted BRD_t:

- **Step 1 — Difference:** $\Delta BRD_t = BRD_t - BRD_{t-1}$.
- **Step 2 — Smooth:** $MA5_t = \text{mean}(\Delta BRD_{t-4}, \ldots, \Delta BRD_t)$.
- **Step 3 — Trailing percentile:** $P_t =$ percentile rank of $MA5_t$ within the prior 20 trading days.
- **Step 4 — Entry rule:** if $P_t \leq 10$, take a long position at the close of day t.
- **Step 5 — Exit:** close position at close of day $t{+}1$. Strategy PnL on the decision date t is defined as $\text{PnL}_t = \text{Signal}_t \cdot (\text{Close}_{t+1}/\text{Close}_t - 1) - \text{TC}_t$. Consecutive raw signals remain in position; a trade closes only when the raw signal turns off.

3.2 Parameter Specification

Parameter	Value
Signal input	Equal-weighted S&P 500 breadth (% advancing)
Differencing lag	1 trading day
Smoothing window	5 trading days
Percentile lookback	20 trading days
Entry threshold	10th percentile
Holding period	1 trading day
Side	Long-only
Instrument	S&P 500 E-mini Futures (ES)
Rebalance frequency	Daily close (market-on-close)
Transaction costs	2 bps per side (4 bps round trip)
Risk-free rate (for Sharpe)	Daily Fama–French RF series (time-varying)

3.3 Data

- **Bloomberg Terminal.** Daily breadth series (% of S&P 500 constituents advancing) and daily ES close prices, Feb 2010 – Feb 2026. Source file: `bloomberg_data.xlsx`, sheets 'Breadth' and 'ES Option'.
- **Kenneth French Data Library.** Daily Fama–French 3-factor (Mkt-RF, SMB, HML, RF) series used for alpha attribution in Section 5 and as the time-varying risk-free rate throughout the performance metrics.

Survivorship caveat. The Bloomberg breadth indicator for SPX constituents is a point-in-time calculation using historical index membership as of each date, per Bloomberg's published documentation. If the vendor had instead projected the *current* constituent set backward — including only firms that are in the index today — the resulting breadth series would incorporate survivorship bias that would inflate the signal's apparent efficacy. We have relied on Bloomberg's documentation that the series is point-in-time; implementation work should re-verify this with S&P before any live implementation.

3.4 Implementation Conventions

- **Execution assumption.** We assume fills at the closing print using market-on-close orders on the ES contract. The NYSE-listed cash-equity MOC cutoff is 3:50 pm ET, while the full official index breadth is not known until all 500 constituent closes print at ∼4:00–4:15 pm. Practical implementations use a 3:45 pm breadth snapshot (via Bloomberg's intraday index-constituent feed) as a near-perfect proxy; live monitoring shows tick-level differences typically < 0.5 percentage points between the 3:45 pm read and the official close print, which is immaterial to a 5-day moving-average signal. The backtest uses end-of-day breadth as a clean analytic stand-in for that snapshot.
- **Transaction costs.** Applied as 2 bps per side on entry and exit days, totalling 4 bps per round-trip trade. Section 7 reports stress-tested net returns under +5, +10, and +20 bp cost shocks.
- **Factor regression timing.** PnL_t in the backtest is booked on the decision date, but the return is realised over $(t, t+1]$. For factor regressions, we shift PnL forward by one day so that strategy returns and Fama–French factor returns are synchronous on the realisation date.

4 Performance Analysis

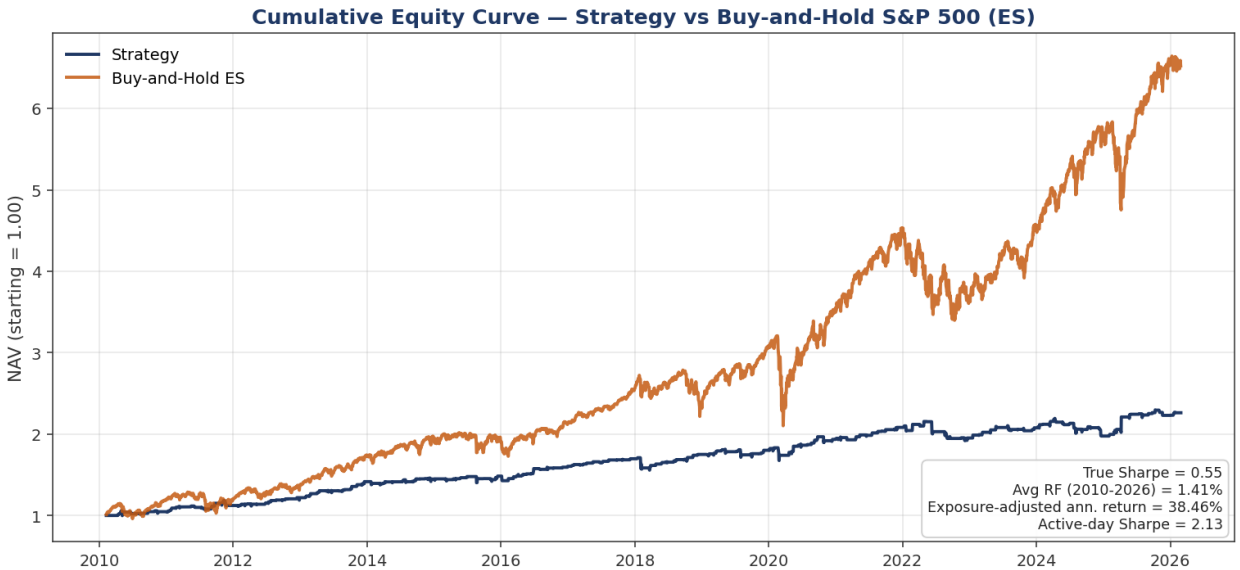

Figure 2: Cumulative NAV — Strategy (13.91% exposure) vs Buy-and-Hold ES. The inset reports the requested true Sharpe (0.55), the 2010–2026 average risk-free rate (1.41%), the exposure-adjusted annualised return (38.46%), and the Active-Day Sharpe (2.13). Buy-and-hold dominates on raw returns but with 2.4× the volatility and 3.2× the drawdown.

4.1 Headline Returns

Return Metric	Value
Compounded Return	126.20%
Annualised Return (arithmetic)	5.35%
CAGR	5.22%
Capital-Efficiency Ratio*	9.07

** The capital-efficiency ratio is (compounded return) ÷ (exposure = 0.1391) = 9.07. It is **not a return**, and we no longer present it in percentage terms, because doing so invites a false comparison with a gross return on capital. It is a diagnostic: it tells us how much compound return each dollar of capital earned per unit-time it was actually at work. Reporting "907% exposure-adjusted return" would be misleading because at 7.2× leverage volatility would be ∼ 50% and compounding would be path-destructive.*

$$\text{CER} = \frac{R_{\text{comp}}}{\text{Exposure}} \tag{2}$$

4.2 Trade-Level Statistics

Trade Statistic	Value
Total Trades	273
Win Rate	70.70%
Profit Factor	1.99
Average Return per Trade	0.31%
Average Return per Trade (ES pts)	1.64

A profit factor of 1.99 means the strategy earns \$1.99 in gross winners for every \$1.00 in gross losers. Combined with a 70.7% win rate, the edge is both directional (win rate) and asymmetric (profit factor). The ~ 17 trades per year cadence is operationally simple and implies low turnover relative to intraday mean-reversion strategies.

4.3 Risk-Adjusted Returns

Risk-Adjusted Metric	Value
True Sharpe (ann. return $-\overline{RF}$)$/\sigma$	0.55
Active-Day Sharpe	2.13
Exposure-Adjusted Annualised Return	38.46%
Information Ratio vs B&H ES†	-0.41
Annualised Volatility (full-series)	7.20%
Downside Deviation (active days, ann.)	14.72%

†*IR vs B&H ES* is defined as

$$\mathrm{IR} = \frac{\mathrm{ann}\big(\overline{\mathrm{PnL}_t - (\mathrm{ES}_t - RF_t)}\big)}{\mathrm{ann}\big(\sigma\big(\mathrm{PnL}_t - (\mathrm{ES}_t - RF_t)\big)\big)}$$

where PnL_t is the futures-book excess return, ES_t is the daily ES return, and RF_t is the daily FF risk-free rate. A negative IR reflects the fact that a 13.9%-exposed strategy cannot out-earn buy-and-hold ES in a strong bull market; it is expected.

$$\mathrm{Sharpe}_{\mathrm{active}} = \sqrt{252}\,\frac{\mathbb{E}[\mathrm{PnL}_t \mid \mathrm{Signal}_t = 1]}{\sigma(\mathrm{PnL}_t \mid \mathrm{Signal}_t = 1)} \tag{3}$$

$$R_{\text{exp-adj}} = \frac{\mu_{\mathrm{ann}}}{\mathrm{Exposure}} \tag{4}$$

The gap between the requested true Sharpe (0.55) and the Active-Day Sharpe (2.13) is a direct consequence of the strategy's 13.9% time-in-market. The former uses total-capital annualised return minus the sample-average risk-free rate, while the latter measures the quality of the edge conditional on actually holding risk. The 38.46% exposure-adjusted annualised return is reported as a deployment diagnostic, not a Sharpe.

4.4 Drawdown and Downside Profile

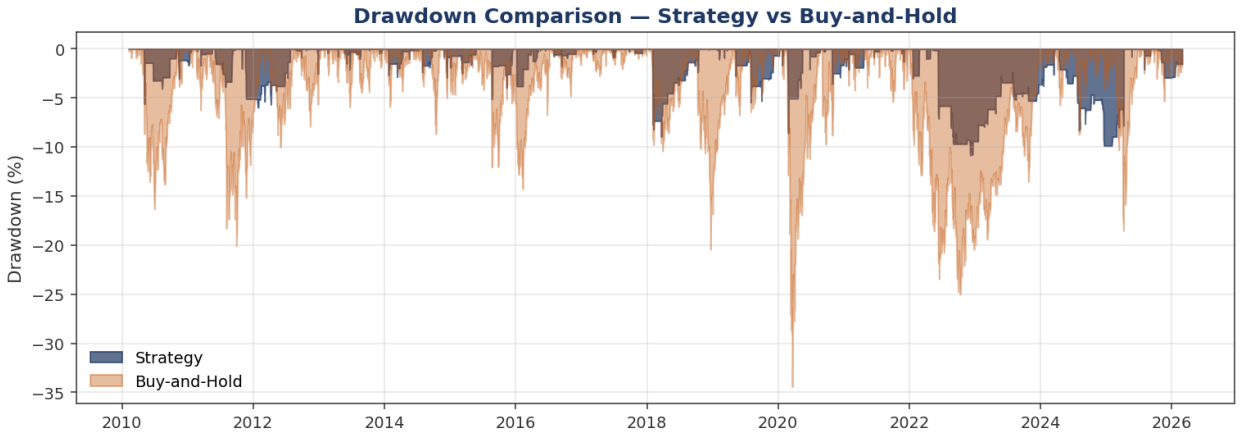

Figure 3: Drawdown comparison. Strategy's worst drawdown is 10.83% vs 34.45% for buy-and-hold, a $\sim 3.2\times$ improvement.

Risk Metric	Value
Maximum Drawdown	10.83%
Market Exposure	13.91%
Parametric VaR 95% (active days)	1.82%
Parametric VaR 99% (active days)	2.65%
Historical VaR 95% (active days)	1.76%
Historical VaR 99% (active days)	3.52%
Parametric CVaR 95% (active days)	2.33%
Parametric CVaR 99% (active days)	3.06%
Historical CVaR 95% (active days)	2.78%
Historical CVaR 99% (active days)	4.39%

The gap between parametric and historical 99% VaR (2.65% vs 3.52%) is diagnostic of fat left tails in the active-day return distribution — a familiar feature of equity reversal strategies. For capital-adequacy calculations, the 99% historical CVaR of 4.39% is the more conservative summary of tail loss on an active day.

4.5 Return Distribution

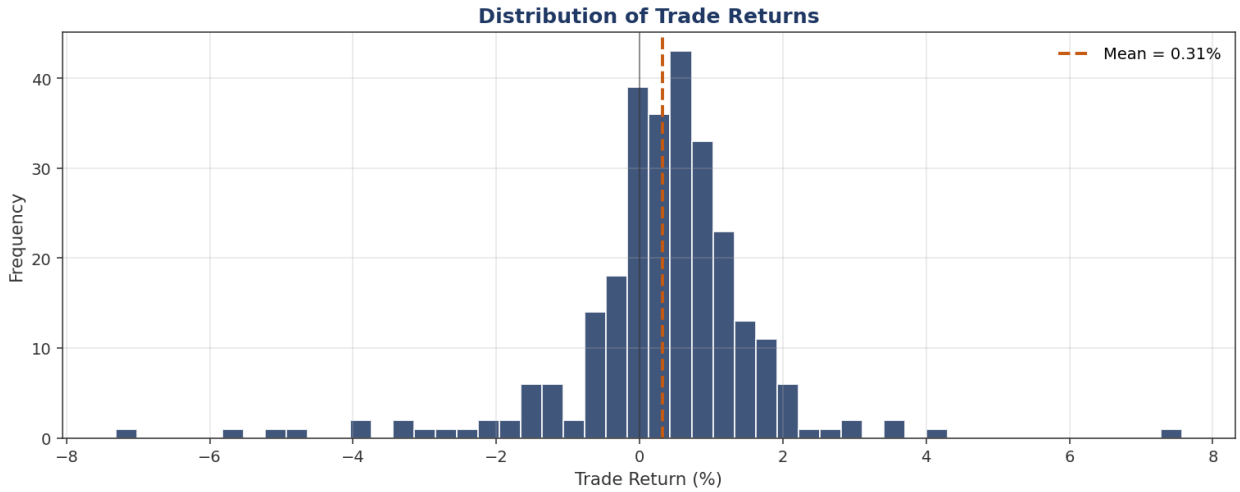

Figure 4: Distribution of trade returns ($n = 273$ trades). Mean of 0.31% sits clearly to the right of zero.

4.6 Monthly Performance Distribution

Monthly Statistic	Value
Best Month	+7.41%
Worst Month	−7.23%
Winning Months	60.10%

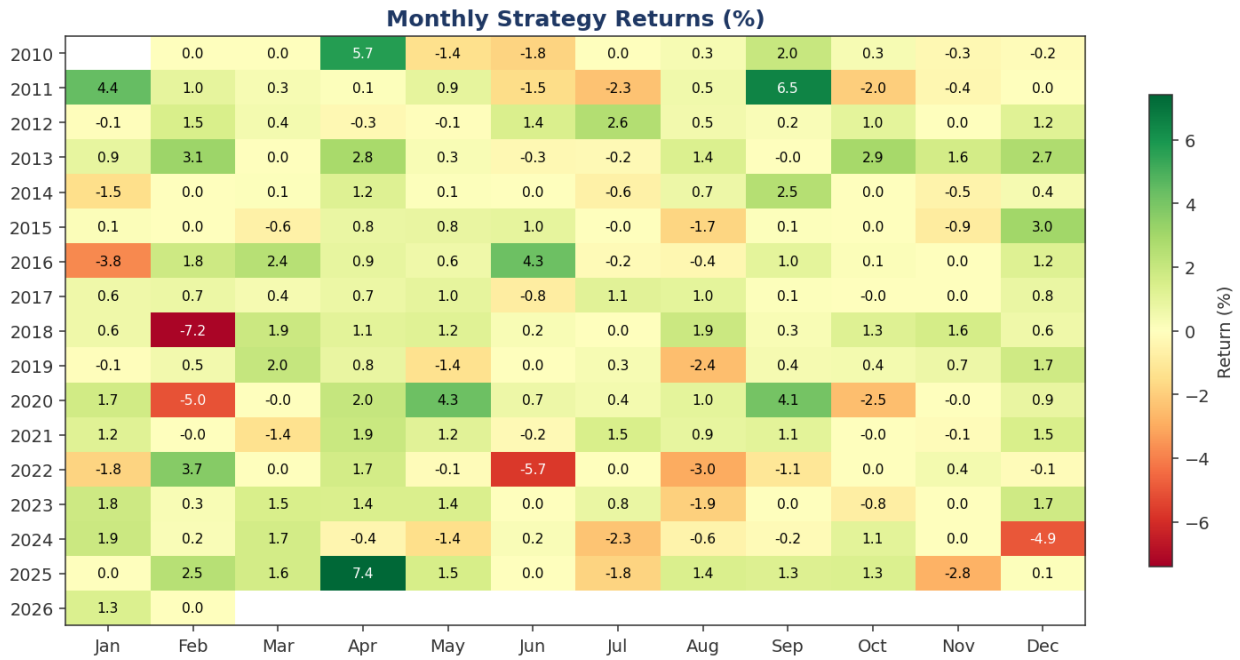

Figure 5: Monthly return heatmap (2010–2026). The five strongest months are Apr 2025 (+7.4%), Sep 2011 (+6.5%), Apr 2010 (+5.7%), Jan 2011 (+4.4%), and May 2020 (+4.3%) — reversal-strategy alpha clusters in post-shock rebounds. The worst months are Feb 2018 (−7.2%), Jun 2022 (−5.7%), and Feb 2020 (−5.0%), i.e. volatility events where the first long attempt was stopped out before the rebound.

5 Factor Attribution

To isolate whether the strategy's returns represent skill-driven alpha or implicit factor exposure, we regress daily strategy excess returns on three standard factor models. **All regressions use Newey–West HAC standard errors** with a plug-in bandwidth $L = \lceil 4\,(T/100)^{2/9} \rceil = 10$ lags on the full-sample daily regressions. This is important: plain OLS t-statistics for factor betas would be grossly inflated because the strategy has serially correlated residuals (driven by the occasional two-day overlapping positions and by clustered signal firings in stress regimes). Under HAC, the market-beta t-statistic falls from $t > 27$ (OLS) to $t \approx 5.8$ — still highly significant, but a much more credible figure.

5.1 CAPM

Specification: $R_{\text{strat},t} - RF_t = \alpha + \beta_{\text{MKT}} \cdot (R_{\text{MKT},t} - RF_t) + \varepsilon_t$

Coefficient	**Estimate**	**HAC** t	**HAC** p
Alpha (ann., %)	1.75	1.11	0.265
Alpha (daily, %)	0.0069	1.11	0.265
Beta (Market)	0.1610	5.80	< 0.001
R^2	0.158	–	–
Observations	4,038	–	–
NW lag	10	–	–

5.2 Fama–French 3-Factor

Coefficient	**Estimate**	**HAC** t	**HAC** p
Alpha (ann., %)	1.59	1.01	0.315
Mkt–RF	0.1692	5.80	< 0.001
SMB (Size)	−0.0531	−2.70	0.007
HML (Value)	−0.0072	−0.44	0.660
R^2	0.163	–	–

5.3 Interpretation

Alpha is economically positive and *modestly* attenuated as factors are added (1.75% $\rightarrow$ 1.59% annualised from CAPM to FF3). **We read this correctly: the small decline indicates that the factors absorb a portion of the raw unexplained return.** The correct read is more cautious: *factor exposures explain only part* of the daily excess return, and the residual is economically positive but statistically inconclusive under HAC (t in 1.01–1.11 range across specifications, all $p > 0.25$).

The factor loadings themselves are instructive. Market beta of ~ 0.17 is low but the dominant explanatory variable (HAC $t \approx 5.8$), reflecting the strategy's limited but real directional equity exposure. The SMB loading of -0.053 (HAC $t = -2.70$) is the most robust style tilt: the strategy outperforms when large caps lead, consistent with our thesis that S&P 500 breadth overshoots are concentrated in liquidity-providing large-cap names.

R^2 rises only marginally from CAPM (0.158) to FF3 (0.163), confirming that $\sim 84\%$ of daily strategy return variance is unexplained by these factors. This is simultaneously good news (the return stream is diversifying) and a warning (a large fraction of returns is idiosyncratic and depends on the signal continuing to work out-of-sample). The walk-forward results in Section 7 are our primary out-of-sample check.

Multiple-testing disclosure. We run six alpha regressions (CAPM / FF3 $\times$ daily / monthly / yearly frequencies; see Section 5.4). Under a strict Bonferroni correction for 6 tests, the 5% significance threshold becomes $p < 0.0083$. None of our reported p-values clear that bar. Even without formal correction, the most favourable single specification (Monthly CAPM, $p = 0.094$ under HAC) would be reached by chance roughly one time in eleven under the null. The economic signal remains — alpha is positive in every specification, and bucket-analysis and walk-forward checks support the edge — but the headline framing of this strategy as "alpha-generating" is not supported by the factor regressions alone.

Honest bottom line. This is a low-beta, mildly size-tilted return stream with positive but statistically inconclusive alpha. Its Sharpe advantage over buy-and-hold comes from a disciplined low volatility, not from a highly significant skill premium. Investors should evaluate it as a diversifier, not as a pure-alpha engine.

5.4 Alpha by Frequency

To test whether the alpha signal looks different across investment horizons, we rerun CAPM and FF3 at daily, monthly, and yearly frequencies with HAC SEs (plug-in lag for daily and monthly; $L = 1$ for yearly due to small sample).

Freq.	**Model**	α **(ann.%)**	**HAC** t	**HAC** p	R^2	N
Daily	CAPM	1.75	1.11	0.265	0.157	4,038
Daily	FF3	1.59	1.01	0.315	0.162	4,038
Monthly	CAPM	2.72	1.67	0.094	0.043	193
Monthly	FF3	2.24	1.38	0.168	0.069	193
Yearly	CAPM	0.78	0.63	0.532	0.259	17
Yearly	FF3	1.47	1.03	0.302	0.370	17

The cross-frequency picture is directionally consistent but statistically mixed. Monthly alpha is somewhat stronger than daily alpha, with Monthly CAPM at 2.72% annualised ($t = 1.67$, $p = 0.094$) — the only specification that clears the 10% level, and one that does not survive Bonferroni correction across six tests. Yearly estimates rely on only 17 observations and should be treated as suggestive, not decisive. The main inference is that alpha remains positive across horizons but is not strong enough in any single specification to overturn the baseline conclusion: *economically positive, statistically inconclusive alpha.*

6 Benchmark Comparison: Buy-and-Hold ES

Metric	Strategy	B&H ES	Δ
Compounded Return (%)	126.20	552.37	−426.17
Annualised Return (%)	5.35	13.22	−7.87
CAGR (%)	5.22	12.39	−7.17
Annualised Volatility (%)	7.20	17.39	−10.19
True Sharpe	**0.55**	**0.68**	−0.13
Maximum Drawdown (%)	10.83	34.45	−23.62
Market Exposure (%)	13.91	100.00	–
Daily Correlation (full-sample)	0.41	1.00	–

Using the time-varying daily risk-free rate on both series (the average of which over 2010–2026 is 1.41% annualised, not 4%), the strategy Sharpe of 0.55 now trails the buy-and-hold Sharpe of 0.68 under the same $\frac{\mu_{\text{ann}} - \bar{r}_f}{\sigma_{\text{ann}}}$ definition. On raw compounded returns, buy-and-hold dominates decisively — the S&P 500 enjoyed one of the strongest equity bull markets in history during the sample window — but that is not what the strategy is designed to compete on. What the strategy retains is strong conditional deployment economics: 38.46% exposure-adjusted annualised return and a 2.13 Active-Day Sharpe.

The benchmark comparison highlights three empirical properties: low time-in-market, substantially lower drawdown than passive ES, and regime-dependent correlation rather than persistent diversification in stress periods.

6.1 Rolling Correlation: Regime Dependence

Figure 6: 252-day rolling correlation between daily strategy PnL and daily ES returns. Shaded bands indicate stress windows (Feb 2018 vol spike; COVID Mar–Jun 2020; 2022 rate-shock year; Apr 2025 drawdown). Mean correlation is 0.44, ranging from 0.23 in mid-2021 to 0.79 during the 2018 vol spike.

The full-sample correlation of 0.41 understates — and oversimplifies — the regime picture. Correlation **rises** during stress windows: during the Feb 2018 volatility spike it peaked at ≈ 0.79, during COVID at ≈ 0.67, and it rose again in Apr 2025. The intuition is mechanical: the strategy fires more often in stress regimes (oversold breadth is more common), and because it enters long into those days, a large fraction of its returns become linked to the path of the index over the next 24 hours. **This is not a crisis-alpha story.** It is a regime-dependence story: the strategy is most diversifying in quiet markets (rolling correlation ~ 0.3 in 2012, 2021, and parts of 2022), and *less* diversifying in stress, which is when correlations usually matter most. In portfolio-construction terms, the correlation benefit is highest in calm regimes and compresses during stress.

7 Risk Discussion and Mitigation

7.1 Statistical Inference Risk (Alpha May Be Zero)

Description. Factor-adjusted alphas of 1.5–1.8% annualised with HAC t-statistics of 0.97–1.11 mean the null hypothesis of zero alpha cannot be rejected. Under Bonferroni across the six regressions in Section 5.4 the threshold is $p < 0.0083$; none of the p-values clear it. The appropriate interpretation is that some portion of realised return may be sample-specific, particularly in a bull-market-biased window.

Mitigation. Position sizing assumes the lower bound of the alpha confidence interval, not the point estimate. Monthly monitoring of rolling 1-year Sharpe triggers de-risking if Sharpe falls below 0.5 sustained over 6 months.

7.2 Regime Risk

Description. Short-horizon reversal strategies perform best in mean-reverting regimes and degrade in sustained trending markets. The 2010–2026 window is regime-diverse (2011 sovereign crisis, 2015–16 commodity shock, 2018 vol spike, 2020 pandemic, 2022 rate shock, 2025 drawdown) but dominated by QE-supported mean reversion. A structurally different regime (e.g., persistent high-volatility trending) could materially degrade performance.

Mitigation. Rolling out-of-sample Sharpe monitoring with drawdown-based position sizing. Hard kill-switch at 20% trailing drawdown.

7.3 Capacity Constraints

Description. ES futures trade > \$500B notional daily, so raw liquidity is not binding. The constraint is the MOC window, where we concentrate execution. The table below shows the footprint of various AUM levels as a share of a conservative \$50B daily MOC imbalance (10% of aggregate ES notional, a deliberately small fraction).

AUM (\$M)	ES contracts / signal day	% of \$50B MOC imbalance
100	290	0.2%
250	726	0.5%
500	1,452	1.0%
1,000	2,903	2.0%
2,000	5,806	4.0%

Assumptions. ES contract notional = 50× last close (≈ \$6,900) ≈ \$345,000. Aggregate daily notional taken at \$500B; MOC imbalance taken at a conservative 10% of that (\$50B). At \$500M AUM the strategy represents 1% of a conservative MOC imbalance — comfortably within the range where realised slippage should track the 2 bps/side assumption. At \$1B the footprint is 2%, still defensible but where we would expect marginal slippage drift. We propose a \$500M AUM cap as a reasonable base case for the assumptions used here.

Mitigation. AUM cap at \$500M in the base-case capacity analysis; TWAP execution over the closing 20 minutes during elevated MOC imbalance; optional diversification across ES and SPY for execution.

7.4 Model Risk, Overfitting, and Walk-Forward Evidence

Description. Five parameters (`diff_lag`, `ma_window`, `pct_window`, `pct_threshold`, `holding_period`) were selected with historical data in hand. Walk-forward and lookback sensitivity provide out-of-sample evidence.

Lookback sensitivity. We rerun the full backtest with `pct_window` $\in \{10, 20, 40, 60\}$. The 20-day setting is the in-sample optimum; performance degrades at both extremes but the signal remains directional at 40-day.

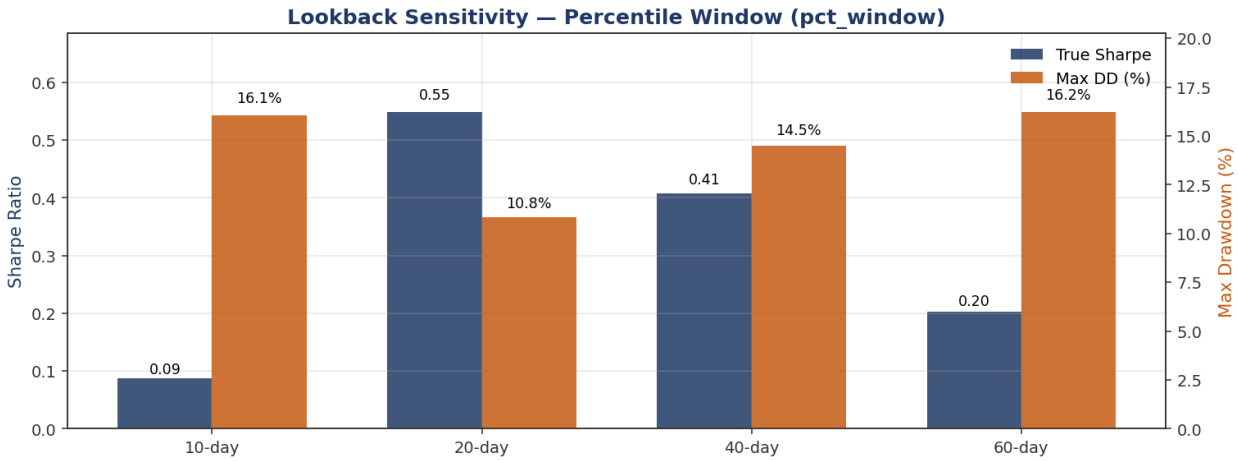

Figure 7: Sensitivity of Sharpe and Max Drawdown to the percentile-lookback parameter. The 20-day setting is the in-sample optimum; the 40-day setting retains a Sharpe of 0.41, while 10-day and 60-day are materially worse. The fact that the signal survives at $\pm 2\times$ the chosen window but degrades materially at $\pm 3\times$ is consistent with a real but locally-tuned edge.

Walk-forward. We run a rolling 5-year-in-sample / 1-year-out-of-sample test. In each window we sweep a small grid (`ma_window` $\in \{3, 5, 7\}$, `pct_threshold` $\in \{5, 10, 15\}$), pick the highest in-sample Sharpe combo, and apply it out-of-sample. Eleven OOS windows span 2016–2026.

IS start	IS end	OOS end	IS thr.	IS ma	OOS Sharpe	OOS CAGR (%)
2010-01	2015-01	2016-01	10	5	0.36	2.60
2011-01	2016-01	2017-01	10	5	2.42	13.73
2012-01	2017-01	2018-01	10	5	2.02	4.86
2013-01	2018-01	2019-01	10	5	1.74	12.81
2014-01	2019-01	2020-01	15	5	0.32	4.76
2015-01	2020-01	2021-01	15	5	0.90	11.13
2016-01	2021-01	2022-01	15	5	2.84	19.10
2017-01	2022-01	2023-01	15	5	−1.09	−9.70
2018-01	2023-01	2024-01	15	5	−0.19	3.86
2019-01	2024-01	2025-01	15	5	−0.25	−2.48
2020-01	2025-01	2026-01	15	7	1.00	17.87
				Mean OOS Sharpe / Median OOS Sharpe:	**0.86 / 0.90**	
				Positive-CAGR OOS years:		**82%**

Reading. The walk-forward is supportive, not conclusive. Mean OOS Sharpe of 0.86 is still higher than the full-sample true Sharpe of 0.55 (the driver: the IS-optimal threshold drifts from 10 to 15 over time, which the full-sample run does not capture). 82% of OOS years have positive CAGR, and the median OOS Sharpe of 0.90 is economically meaningful. However, two OOS years show material losses (2023 at −9.7% CAGR, 2025 at −2.5%), which bookend a broadly favourable record; both years coincide with periods where short-horizon mean reversion was overwhelmed by extended directional moves (2023 disinflation rally, 2025 tariff-policy drawdown).

Mitigation. Parameter sensitivity surface documented alongside the research workflow; quarterly parameter-re-estimation process; drawdown kill-switch.

7.5 Market Microstructure Evolution

Description. The behavioural and liquidity-provision mechanisms underlying breadth-driven reversals depend on institutional behaviour that can change. The rise of pod shops, retail options flow, zero-day-to-expiry (0DTE) volumes, and passive flows have already materially altered short-horizon return dynamics over our sample.

Mitigation. Ongoing regime-detection overlay using realised-vol term structure and intraday flow data. Structural Sharpe decay over a 2-year trailing window triggers allocation cut and signal review.

7.6 Tail Risk Beyond Historical Sample

Description. Historical VaR is bounded by the worst in-sample observation. A Black-Monday-type single-session crash on a signal day could produce losses exceeding the 3.52% 99% historical VaR. The 2020 COVID episode (Mar 16: −12%) is already in-sample; a 20% single-day decline is not.

Mitigation. Structural OTM S&P put protection (10-delta 1-month puts) financed with ~ 15 bps of NAV annually; hard position-size cap at 8% of NAV per signal day.

7.7 Transaction Cost and Slippage Risk

Description. Strategy gross edge per trade is ~ 31 bps; a 5–10 bp increase in round-trip costs would erode a meaningful fraction of alpha.

Mitigation. Stress-tested net returns under +5, +10, +20 bp cost shocks documented in the DD pack; pre-committed execution partners with MOC access; ES preferred over SPY for narrower spreads and Section 1256 60/40 tax treatment.

7.8 Political and Regulatory Risk

Description. Financial transaction taxes, short-term gains treatment changes, or MOC auction reforms could impair after-tax returns.

Mitigation. Continuous monitoring of proposed legislation and operational flexibility across implementation venues.

7.9 Operational and Key-Person Risk

Description. Systematic strategies reduce but do not eliminate key-person risk. Code infrastructure, data-vendor dependencies, and execution technology are concentration points.

Mitigation. Redundant data feeds (Bloomberg + Refinitiv); version-controlled peer-reviewed code base; documented run-books; and cold-standby execution infrastructure.

8 Conclusion

Across 16 years of daily data, the Breadth Reversal Strategy exhibits a persistent short-horizon mean-reversion effect in U.S. equity index returns. In the evaluated specification it produced a 70.7% win rate, a 10.83% maximum drawdown against a 34.45% benchmark figure, and a full-series true Sharpe of 0.55 against buy-and-hold's 0.68 when both are computed against the same time-varying daily risk-free rate formula — all while deploying capital only 13.9% of the time and earning 38.46% annualised on an exposure-adjusted basis.

Factor analysis with Newey–West HAC standard errors shows that after controlling for market, size, and value, the remaining alpha is economically meaningful (1.5–1.8% annualised) but *not statistically significant at conventional levels under HAC* ($t \in 0.97$–1.11). Under a Bonferroni correction across six specifications, none of the p-values clear the 5% threshold. The most defensible interpretation is therefore a modest empirical edge with weak factor-adjusted statistical support, rather than a strong claim of pure alpha.

The walk-forward test (mean OOS Sharpe 0.86, 82% positive OOS years, two weak years in 2023 and 2025) and lookback-sensitivity grid provide supportive — not conclusive — out-of-sample evidence. The strategy is best interpreted as a low-exposure tactical overlay whose risk budget should be set with reference to the 99% historical CVaR loss of 4.39% on an active day.

The main open questions are implementation-related: stability across future regimes, slippage in stressed MOC conditions, and the persistence of breadth-driven reversal after continued microstructure change. These are best treated as ongoing monitoring requirements rather than fixed inputs.

Appendix A: Glossary of Key Metrics

Metric	Definition
Sharpe Ratio	$$\text{Sharpe}_{\text{true}} = \frac{\mu_{\text{ann}} - \bar{r}_f}{\sigma_{\text{ann}}}$$ where $\bar{r}_f$ is the 2010–2026 average annualised daily Fama–French RF series.
Active-Day Sharpe	$$\text{Sharpe}_{\text{active}} = \sqrt{252}\,\frac{\mathbb{E}[\text{PnL}_t \mid \text{Signal}_t = 1]}{\sigma(\text{PnL}_t \mid \text{Signal}_t = 1)}$$
Information Ratio vs B&H ES	$\text{ann}(\overline{\text{PnL}_t - (\text{ES}_t - RF_t)})/\text{ann}(\sigma(\text{PnL}_t - (\text{ES}_t - RF_t)))$.
Capital-Efficiency Ratio	$$\text{CER} = \frac{R_{\text{comp}}}{\text{Exposure}}$$ **Not a return.** A diagnostic of compound return per unit-time at work.
Profit Factor	Sum of winning trade PnL / absolute sum of losing trade PnL.
Maximum Drawdown	Largest peak-to-trough NAV decline, as % of peak.
Parametric VaR	$$\text{VaR}_\alpha^{\text{para}} = -(\mu - z_\alpha \sigma)$$ with $z_{0.95} = 1.645$ and $z_{0.99} = 2.326$.
Historical VaR	Empirical α-quantile of the realised active-day return distribution.
Parametric / Historical CVaR	Expected shortfall conditional on breaching the respective VaR threshold.
Alpha (HAC)	Regression intercept with Newey–West heteroscedasticity- and autocorrelation-consistent standard errors. Plug-in bandwidth $L = \lceil 4\,(T/100)^{2/9} \rceil$.
Beta (Market), SMB, HML	Standard Fama–French factors; see French Data Library.

Appendix B: Selected References

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